What fills the context window

Every call re-sends the whole thing, and all of it competes for one ceiling.

FIELD GUIDE TO THE AI STACK
mchelen.github.io/FieldGuideAIStack

COMPONENT · PROCESS · TECHNIQUE · CONTROL · CEILING · ARTIFACT · STORE

THE WINDOW

a per-call ceiling on all of it together — not a budget that refills

01 System prompt

the vendor's standing instructions

Role, rules and format, sent ahead of the conversation on every request. It is the part of the product you cannot read or edit.

YOU NEVER SEE IT

02 Tool definitions

what it is allowed to ask for

Every function's name, description and arguments. They are part of the prompt, so a large toolset is a standing cost on the window.

CHARGED WHETHER USED OR NOT

03 The conversation

there is nowhere else for it to live

Not a store the agent writes to — the transcript, re-sent whole. This is the part that runs you out of room.

GROWS EVERY TURN

Compaction
Older turns replaced by a summary so a long run keeps fitting.

Session
Ends, and takes the window with it. A fresh one often beats arguing.

04 Retrieved material

fetched at question time

Documents pulled in for this question, so the answer rests on them rather than on what the model absorbed in training.

THE POINT IS GROUNDING

Chunking
Documents split small enough to retrieve a piece rather than the whole.

Semantic search
Retrieval by meaning rather than by matching words.

Reranking
A second pass that reorders hits before any of them cost tokens.

Grounding
Answering from the retrieved source instead of from memory.

THE UNIT IT IS MEASURED IN

Token — A word fragment. The window, the bill and the rate limits are all counted in these, not in words.

THE WORK OF DECIDING

Context engineering — What goes in, what gets dropped, what gets fetched again — most of what people call prompt engineering is this.

WHAT IT IS NOT

Long-term memory — State across sessions is a directory the harness reads back into the window. The window itself remembers nothing.

Nothing in the window persists. Every call re-sends all of it, the ceiling applies to the whole, and what looks like memory is a file something re-read for you.